# Class 18: Pertussis Mini Project

Linh Dang (PID: A16897764)

## Background

Pertussis (a.k.a whooping cough) is a common lung infection caused by the bacteria *B. Pertussis*. The CDC tracks cases of Pertussis in the US: https://tinyurl.com/pertussiscdc

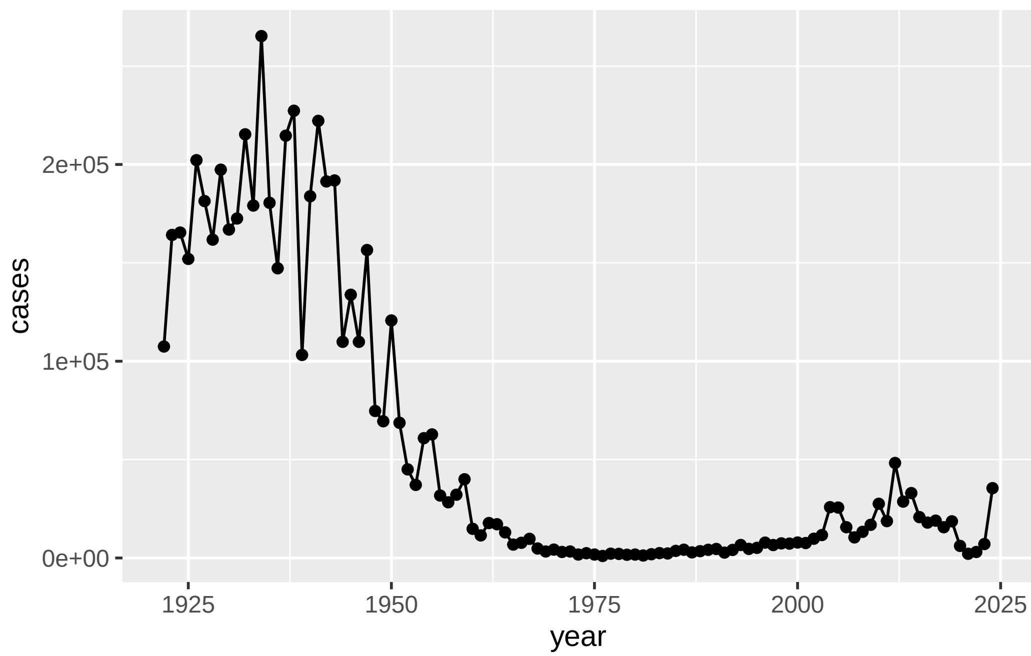
## Examing cases of pertussis by year

We can use the **datapasta** package to scrape case numbers from the CDC

```
cdc <- data.frame(year = c(1922L,1923L,1924L,1925L,1926L,1927L,1928L,1929L,
1930L,1931L,1932L,1933L,1934L,1935L,1936L,1937L,1938L,1939L,1940L,1941L,1942L,1943L,1944L,194
1970L,1971L,1972L,1973L,1974L,1975L,1976L,1977L,1978L,1979L,1980L,1981L,1982L,1983L,1984L,198
2010L,2011L,2012L,2013L,2014L,2015L,2016L,2017L,2018L,2019L,2020L,2021L,2022L,2023L,2024L),
            cases = c(107473,164191,165418,152003,202210,181411,161799,197371,166914,172559,2
          )
```

> Q1. Make a plot of Pertussis cases per year using ggplot?

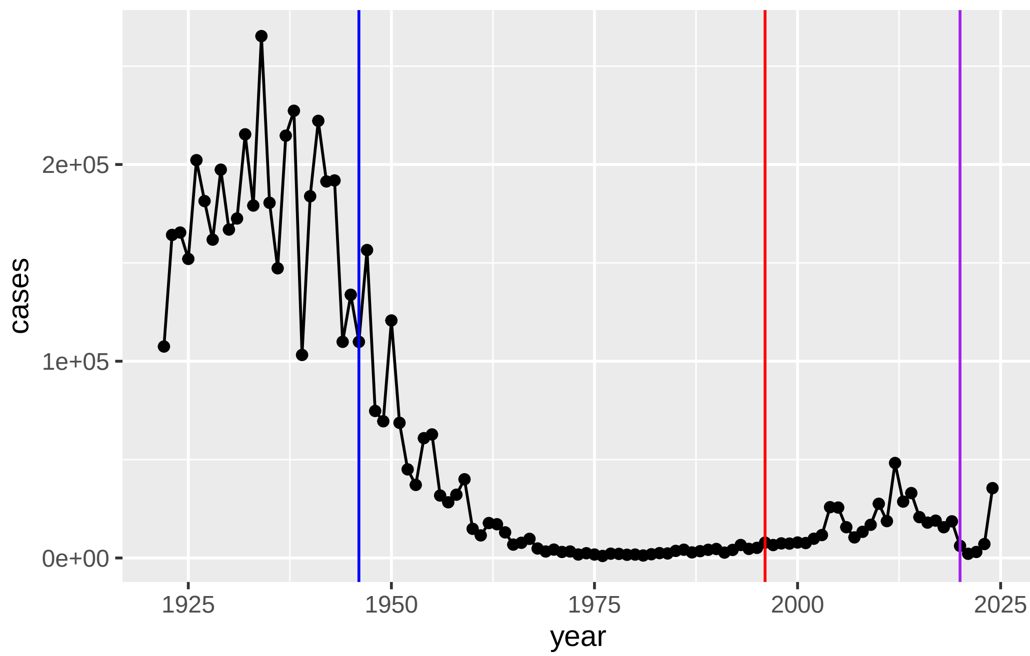
```
library(ggplot2)

cases <- ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line()
cases
```




Q2. Add some key time points in our history of interaction with Pertussis. These include wP roll-out (the first vaccine) in 1945 and the switch to aP in 1996?

We can use `geom_vline()` for this

```
cases +
  geom_vline(xintercept = 1946, col="blue") +
  geom_vline(xintercept = 1996, col="red") +
  geom_vline(xintercept = 2020, col="purple")
```




The number of Pertussis decrease significantly from more than 20,000 cases to half amount after 1945 and the number remain relatively low for more than 50 years. Until COVID happened, Pertussis cases are rising again.

Mounting evidence suggests that the newer **aP** vaccine is less effective over long term than the older **wP** vaccine that it replaced. In other words, vaccine protection wanes more rapidly with aP than with wP.

## Enter the CMI-PB project

CMI-PB (Computational Models of Immunity - Pertussis Boost). Major goal is to investigate how the immune system responds differently to with aP versus wP vaccinated individuals and be able to predict this at an early time.

CMI_PB makes all their collected data freely available and they store it in a database composed different tables. Here we will access a few of these.

We can use the **jsonlite** package to read this data

```
library(jsonlite)

subject <- read_json("https://www.cmi-pb.org/api/v5_1/subject",
                     simplifyVector = TRUE)
```

```r
head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

> Q3. How many subjects (i.e. enrolled people) are there in this dataset?

```r
nrow(subject)
```

```
[1] 172
```

> Q4. How many "aP" and "wP" subjects are there?

```r
table(subject$infancy_vac)
```

```

aP wP 
87 85 
```

> Q5. How many Male/Female are in the dataset?

```r
table(subject$biological_sex)
```

```

Female   Male 
   112     60 
```

> Q6. How about gender and race numbers?

```r
table(subject$race, subject$biological_sex)
```

```
                                            Female Male
  American Indian/Alaska Native                  0    1
  Asian                                         32   12
  Black or African American                      2    3
  More Than One Race                            15    4
  Native Hawaiian or Other Pacific Islander      1    1
  Unknown or Not Reported                       14    7
  White                                         48   32
```

Q7. Is this representative of the US population?

No

Let's read another database table from CMI-PB

```r
specimen <- read_json("https://cmi-pb.org/api/v5_1/specimen",
                      simplifyVector = TRUE)
ab_data <- read_json("https://cmi-pb.org/api/v5_1/plasma_ab_titer",
                     simplifyVector = TRUE)
```

Wee peak at these

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

We want to “join” these tables to get all our information together. For this we will use **dplyr** package and the `inner_join()` function.

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
```

```
4                            7                            7          Blood
5                           11                           14          Blood
6                           32                           30          Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```r
head(ab_data)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

One more "join" to get ab_data anad meta all together

```r
abdata <- inner_join(ab_data, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
```

```
4           1     IgG               TRUE     PRN  332.12718       2.602350
5           1     IgG               TRUE     FHA 1887.12263      34.050956
6           1     IgE               TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

```r
dim(abdata)
```

```
[1] 61956    20
```

Q. How many Ab isotypes are there in the dataset?

```r
table(abdata$isotype)
```

```

  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q. How many different antigens are measured in the dataset?

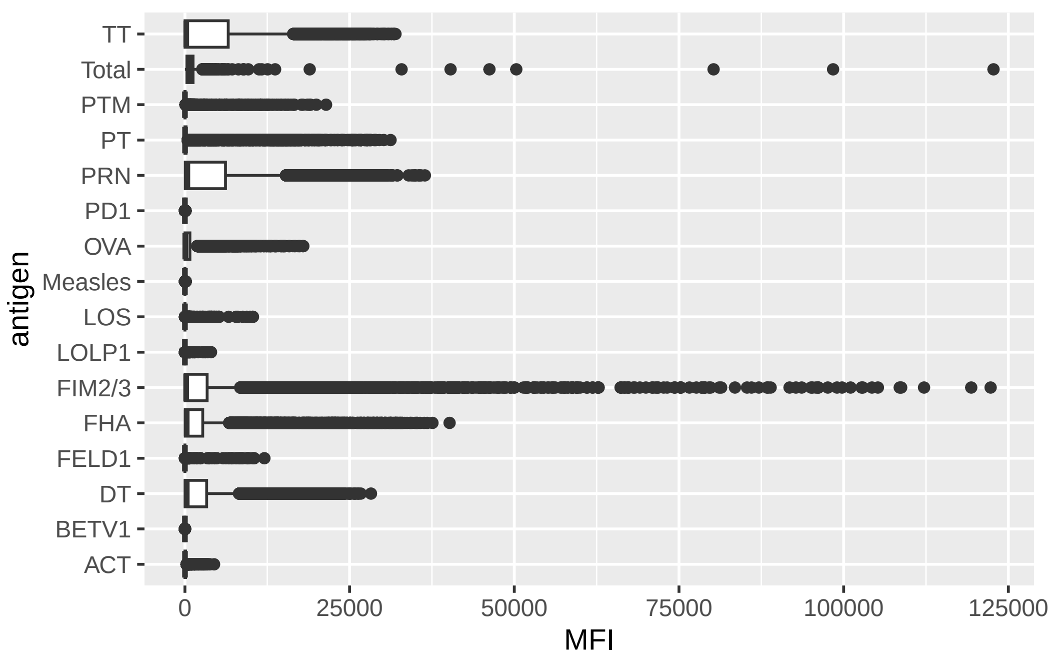
```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

Q. Make a boxplot of antigen leveles accross the whole dataset (MFI vs anigen)?

```r
ggplot(abdata) +
  aes(MFI, antigen) +
  geom_boxplot()
```

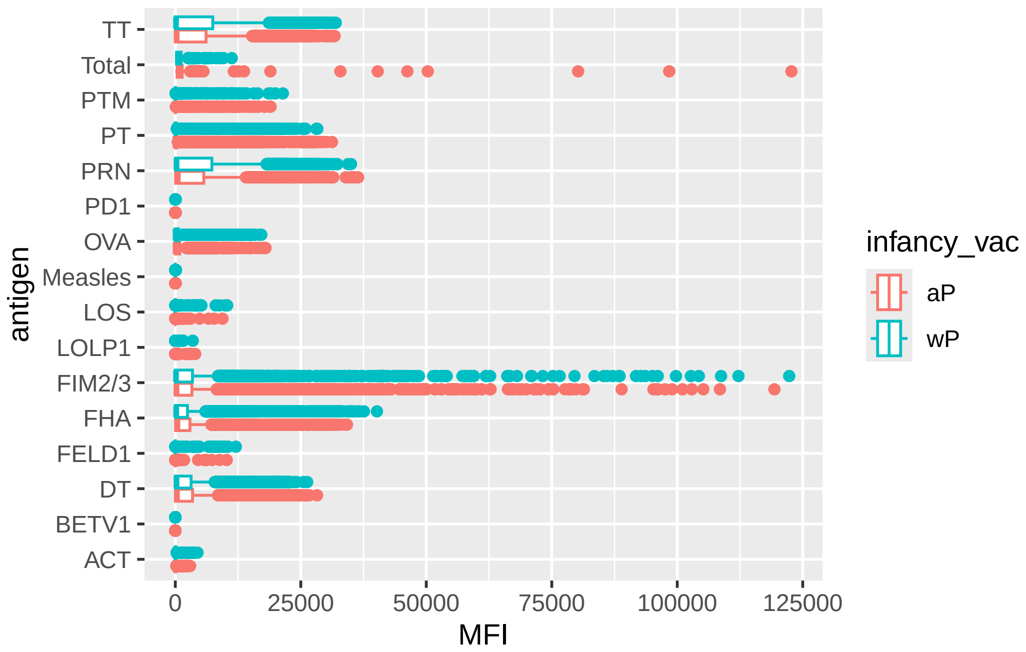
```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_boxplot()`).
```




Q. Are there obvious different between aP and wP values/

```r
ggplot(abdata) +
  aes(MFI, antigen, col=infancy_vac) +
  geom_boxplot()
```

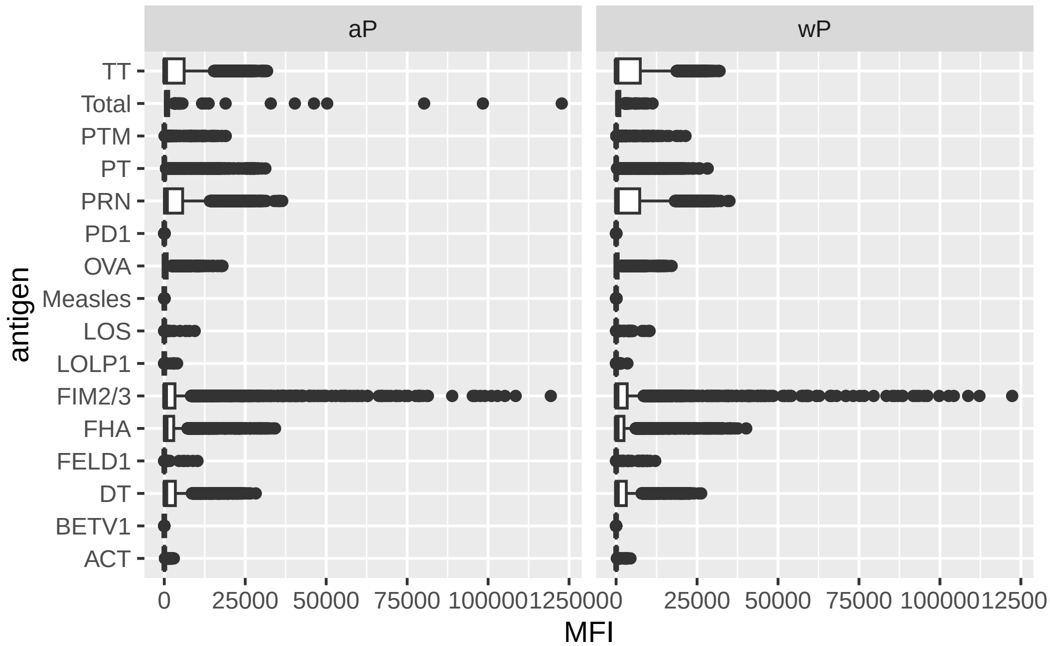
```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_boxplot()`).
```




Or we can "facet" by `infancy_vac` to get two individuals plots one for each value of `infancy_vac`.

```r
ggplot(abdata) +
  aes(MFI, antigen) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_boxplot()`).
```




### Focus on IgG levels

IgG is the most abundant antibody in blood. With four sub classes (IgG1 to IgG4). for long-term immunity

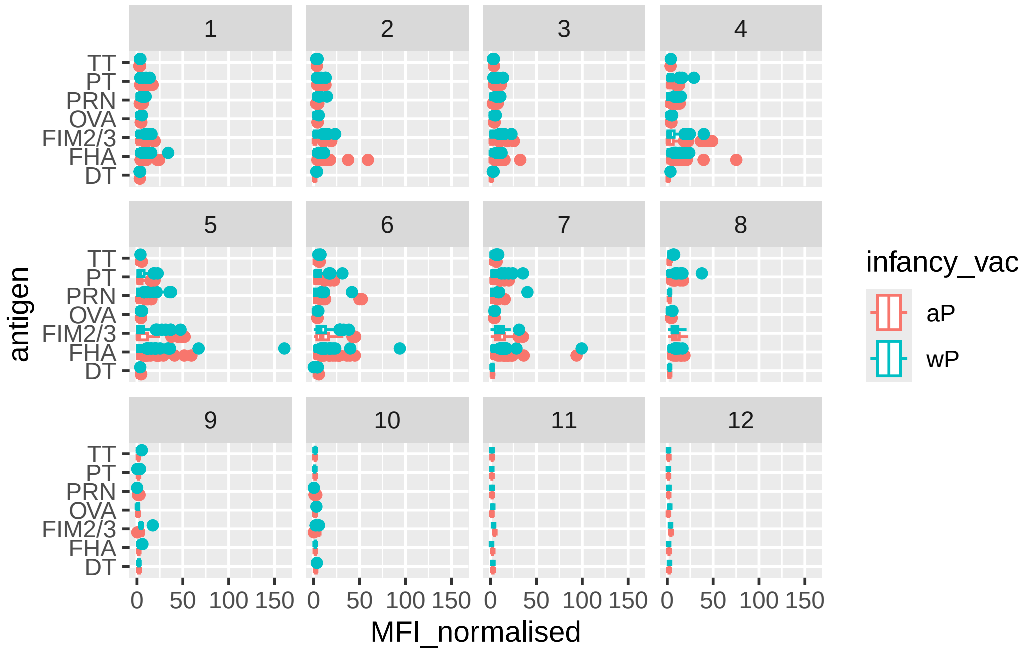
```r
igg <- abdata |>
  filter(isotype == "IgG")
head(igg)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
```

```
6 IU/ML                 4.679535          3          wP         Female
                ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Same boxplot of antigens as before

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(~visit)
```




Focus in further in just one of these antigens - let's pick **PT** (Pertussis toxin, one of the main toxins of the bacteria) in the **2021_dataset** again for **IgG** antibody isotopes

```
table(igg$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
        1182         1617         1456         3010
```

```
pt_igg <-abdata |>
  filter(isotype=="IgG",
         antigen=="PT",
         dataset=="2021_dataset")
```

```
dim(pt_igg)
```

```
[1] 231  20
```

```
ggplot(pt_igg) +
  aes(actual_day_relative_to_boost,
      MFI_normalised,
```

```
    col=infancy_vac,
    group=subject_id) +
  geom_point() +
  geom_line() +
  theme_bw() +
  geom_vline(xintercept = 0) +
  geom_vline(xintercept = 14)
```